

# Soft Spaces - Phase 4

## v40.1 Ranked-Search Noise Robustness - Run Report

Cross-energy ranked search under controlled source-side score noise

**Executive conclusion.** The full 12-seed v40.1 run passed the predeclared primary checkpoint at  $\eta = 1e-3$ . The REAL-specific ranked-search advantage remained statistically significant through  $\eta = 0.4$  on the tested grid and failed only at  $\eta = 0.8$  under the full four-gate decision rule.

### 1. Frozen experimental setup

| Parameter            | Frozen value                            |
|----------------------|-----------------------------------------|
| Phase-3 kernel       | v30.7 / namespace v30.2                 |
| Qubits               | 8                                       |
| Hamiltonian terms    | 11                                      |
| Independent seeds    | 25043000 - 25043011                     |
| Application          | -E prominence used to rank search on +E |
| Top-K target set     | K = 8                                   |
| Noise repetitions    | 500 per seed/model/ $\eta$              |
| Bootstrap replicates | 200,000                                 |

**Noise definition.** Source-side prominence was corrupted as  $\text{source\_noisy} = \text{source} + \eta * \text{std}(\text{source}) * \text{Normal}(0, 1)$ . Noise was applied only to the -E information used for ranking. +E target prominence remained untouched and was used only after ranking to score Precision@8, recovery effort, and NDCG@8.

**Important scope restriction.**  $\eta$  is dimensionless relative score noise.  $\eta = 1e-3$  is not a physical two-qubit gate-error probability.

### 2. Noise-sweep results

| $\eta$ | REAL P@8 | NULL P@8 | Delta P   | 95% CI Delta P        | sign p   | Wilcoxon p | REAL eval | NULL eval | Decision |
|--------|----------|----------|-----------|-----------------------|----------|------------|-----------|-----------|----------|
| 0      | 1.000000 | 0.822917 | +0.177083 | [+0.125000,+0.229167] | 0.000488 | 0.000976   | 8.000     | 10.917    | PASS     |
| 1e-4   | 1.000000 | 0.822917 | +0.177083 | [+0.125000,+0.229167] | 0.000488 | 0.000976   | 8.000     | 10.917    | PASS     |
| 3e-4   | 1.000000 | 0.822917 | +0.177083 | [+0.125000,+0.229167] | 0.000488 | 0.000976   | 8.000     | 10.917    | PASS     |
| 1e-3   | 0.998292 | 0.822917 | +0.175375 | [+0.123292,+0.229167] | 0.000488 | 0.000976   | 8.016     | 10.919    | PASS     |
| 3e-3   | 0.994542 | 0.822917 | +0.171625 | [+0.114583,+0.229167] | 0.000488 | 0.000976   | 8.059     | 10.934    | PASS     |
| 0.01   | 0.992521 | 0.824479 | +0.168042 | [+0.113583,+0.225542] | 0.000488 | 0.000976   | 8.082     | 10.953    | PASS     |
| 0.03   | 0.980354 | 0.825521 | +0.154833 | [+0.103729,+0.209708] | 0.000488 | 0.000976   | 8.218     | 10.955    | PASS     |
| 0.1    | 0.947292 | 0.824375 | +0.122917 | [+0.080438,+0.172917] | 0.000488 | 0.000976   | 8.667     | 11.037    | PASS     |
| 0.2    | 0.913500 | 0.815313 | +0.098187 | [+0.056792,+0.150083] | 0.000488 | 0.000976   | 9.282     | 11.325    | PASS     |
| 0.4    | 0.848021 | 0.781458 | +0.066563 | [+0.024187,+0.114625] | 0.032715 | 0.012689   | 10.906    | 12.360    | PASS     |
| 0.8    | 0.737125 | 0.697500 | +0.039625 | [+0.005146,+0.075687] | 0.113281 | 0.027330   | 14.608    | 15.191    | FAIL     |

### 3. Primary predeclared checkpoint: $\eta = 1e-3$

| Metric                 | Result                 |
|------------------------|------------------------|
| Mean Precision@8 REAL  | 0.998292               |
| Mean Precision@8 NULL  | 0.822917               |
| Mean Delta Precision@8 | +0.175375              |
| 95% bootstrap CI       | [+0.123292, +0.229167] |

|                       |            |
|-----------------------|------------|
| One-sided sign-test p | 0.00048828 |
| One-sided Wilcoxon p  | 0.00097609 |

| Gate | Criterion                          | Decision    |
|------|------------------------------------|-------------|
| C1   | Mean Delta Precision@8 > 0         | <b>PASS</b> |
| C2   | 95% bootstrap CI lower bound > 0   | <b>PASS</b> |
| C3   | One-sided exact sign-test p < 0.05 | <b>PASS</b> |
| C4   | One-sided Wilcoxon p < 0.05        | <b>PASS</b> |

## FINAL v40.1 PRIMARY CHECKPOINT DECISION: PASS

### 4. Robustness envelope

The largest tested noise level satisfying all four Precision@8 gates was **eta = 0.4**. At this point REAL retained mean Precision@8 = 0.848021 versus 0.781458 for NULL, with Delta Precision@8 = +0.066563 and a 95% bootstrap interval of [+0.024187, +0.114625]. The full gate failed at eta = 0.8 because the sign-test criterion no longer met p < 0.05.

At eta = 1e-3, REAL required 8.016 target evaluations on average versus 10.919 for NULL, saving 2.902 evaluations. Even at eta = 0.4, REAL still required fewer evaluations (10.906 versus 12.360).

## 5. Seed-level results at $\eta = 1e-3$

| Seed     | REAL P@8 | NULL P@8 | Delta P   | REAL eval | NULL eval | Saved  | REAL NDCG | NULL NDCG | Delta NDCG |
|----------|----------|----------|-----------|-----------|-----------|--------|-----------|-----------|------------|
| 25043000 | 1.000000 | 0.750000 | +0.250000 | 8.000     | 12.000    | +4.000 | 1.000000  | 0.980456  | +0.019544  |
| 25043001 | 1.000000 | 0.750000 | +0.250000 | 8.000     | 12.000    | +4.000 | 1.000000  | 0.690019  | +0.309981  |
| 25043002 | 1.000000 | 1.000000 | +0.000000 | 8.000     | 8.000     | +0.000 | 1.000000  | 0.979418  | +0.020582  |
| 25043003 | 1.000000 | 0.875000 | +0.125000 | 8.000     | 9.000     | +1.000 | 1.000000  | 0.960462  | +0.039538  |
| 25043004 | 1.000000 | 0.875000 | +0.125000 | 8.000     | 10.000    | +2.000 | 1.000000  | 0.980043  | +0.019957  |
| 25043005 | 1.000000 | 0.875000 | +0.125000 | 8.000     | 9.000     | +1.000 | 0.999998  | 0.864639  | +0.135359  |
| 25043006 | 1.000000 | 0.625000 | +0.375000 | 8.000     | 13.000    | +5.000 | 1.000000  | 0.760924  | +0.239076  |
| 25043007 | 1.000000 | 0.875000 | +0.125000 | 8.000     | 11.000    | +3.000 | 1.000000  | 0.945568  | +0.054432  |
| 25043008 | 1.000000 | 0.750000 | +0.250000 | 8.000     | 13.000    | +5.000 | 0.999279  | 0.996623  | +0.002656  |
| 25043009 | 1.000000 | 0.875000 | +0.125000 | 8.000     | 11.000    | +3.000 | 0.999989  | 0.975744  | +0.024245  |
| 25043010 | 0.979500 | 0.875000 | +0.104500 | 8.196     | 9.022     | +0.826 | 0.999867  | 0.877200  | +0.122667  |
| 25043011 | 1.000000 | 0.750000 | +0.250000 | 8.000     | 14.000    | +6.000 | 1.000000  | 0.765639  | +0.234361  |

**Special note on seed 25043002.** This seed has only eight +/-E pairs. With  $K = 8$ , Precision@8 cannot discriminate REAL from NULL because both necessarily include the full candidate set. NDCG remains informative because ranking order still differs.

## 6. Scientific interpretation

v40.1 extends the v40.0 application result by showing that the REAL-specific ranked-search advantage is not confined to the noiseless source ranking. The advantage remains reproducible under substantial corruption of the source-side score and retains statistical support over a broad tested noise range.

The result therefore supports robustness of the application signal itself. It does not yet establish robustness to physical quantum-hardware noise.

## 7. Scope and limitations

- Classical score-noise robustness experiment on the frozen 8Q application model.
- Not a Qiskit circuit simulation.
- Not an IBM Heron or other quantum-hardware experiment.
- $\eta$  is relative score noise, not a gate-error probability.
- No claim of quantum computational advantage is supported by this run.

## 8. Recommended next step

Freeze v40.1 as the score-noise robustness result. The next experiment should introduce a hardware-proxy or explicit circuit-level realization with finite shots, readout error, gate/depolarizing noise, and circuit depth, while preserving the same ranked-search endpoint and REAL-vs-NULL comparison.